

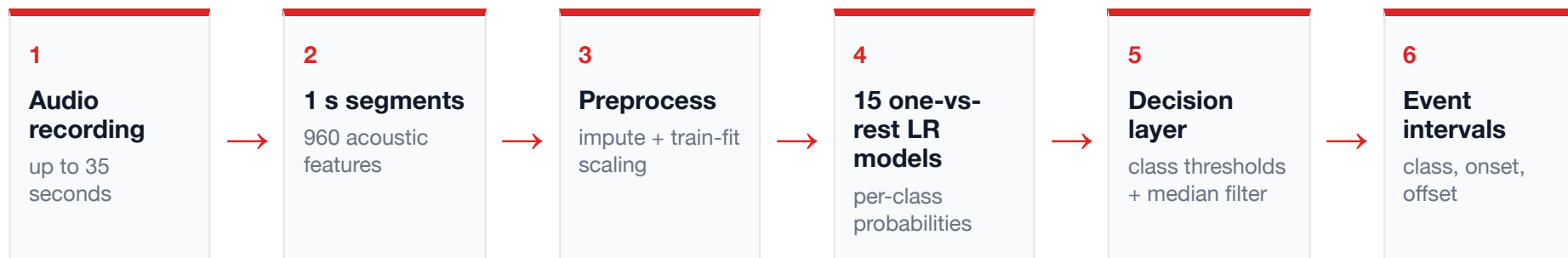
The Challenge

Machine Learning Pattern Classification

Team Quantized Encoders

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Final system: segment-level **sound event detection**



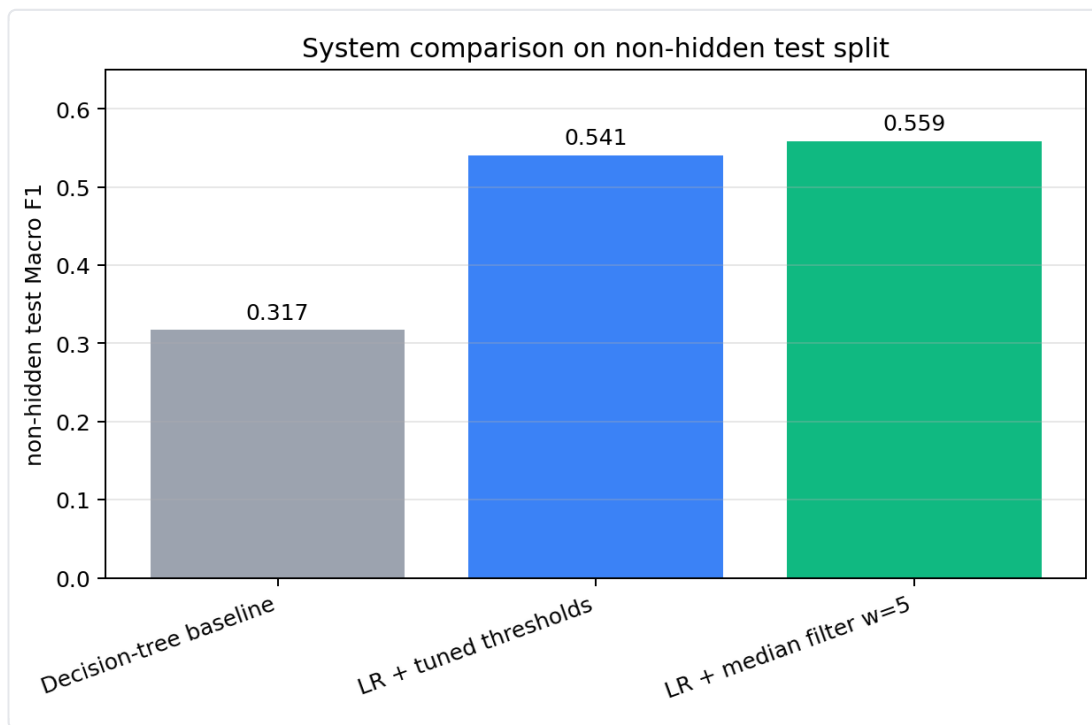
Leakage control

All segments from a recording remain in one split. Scaling, hyperparameters, and thresholds are fitted only on development data.

Evaluation target

Segment-based Macro F1 at 1-second resolution: each class matters equally, including rare transient sounds.

Hypothesis 1: a regularized linear model can beat the tree baseline



OUTCOME

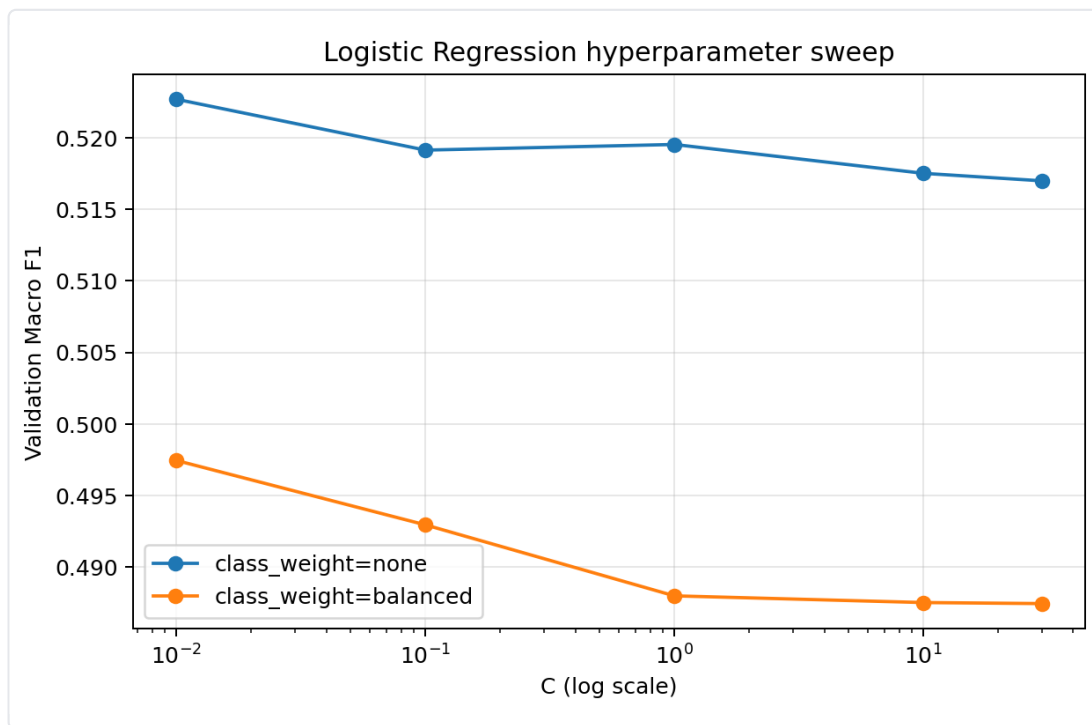
0.317 → 0.541

+22.4 Macro-F1 points on the non-hidden test set after replacing independent decision trees with standardized, one-vs-rest Logistic Regression and tuned decision thresholds.

Largest gains: **keyboard typing** (+0.354), **microwave** (+0.321), and **toilet flushing** (+0.285).

Short, acoustically similar events like window/door and drawer actions remain difficult.

Hypothesis 2: balance the rare classes or calibrate decisions?



VALIDATION SWEEP

Best model: `C = 0.01`, no class weighting.

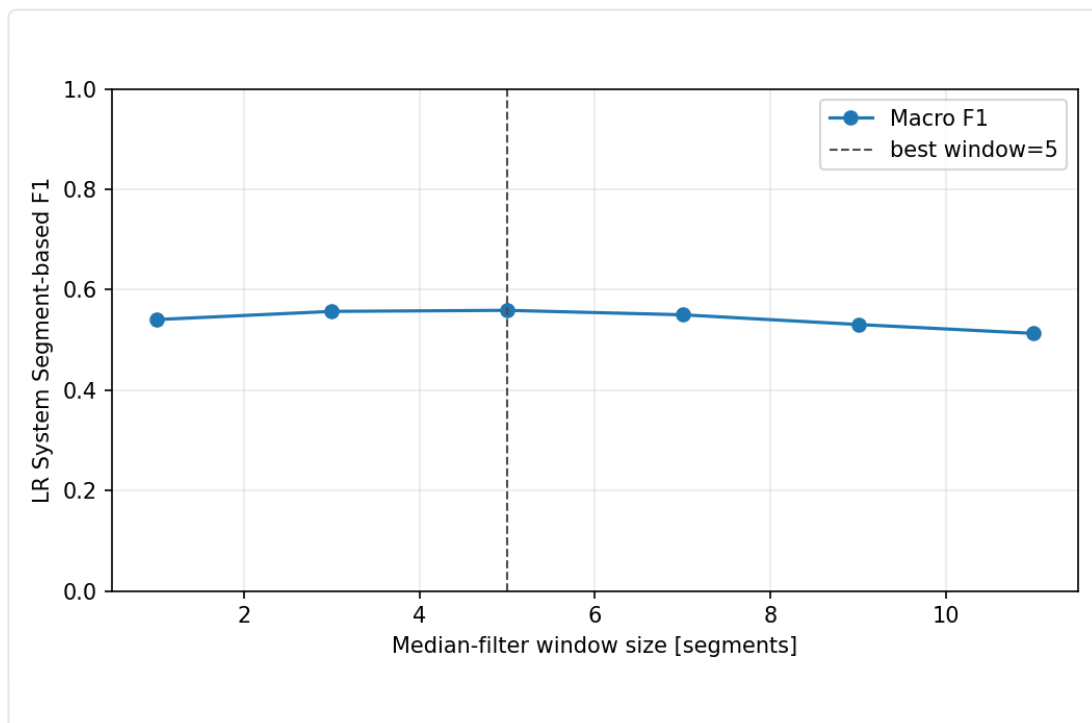
Balanced weighting was rejected

Macro F1 fell from 0.523 to 0.497 at `C = 0.01`.

Per-class thresholds were retained

They let each class trade precision and recall according to its score distribution instead of enforcing 0.5 globally.

Hypothesis 3: nearby segments can correct isolated mistakes



TEMPORAL REFINEMENT

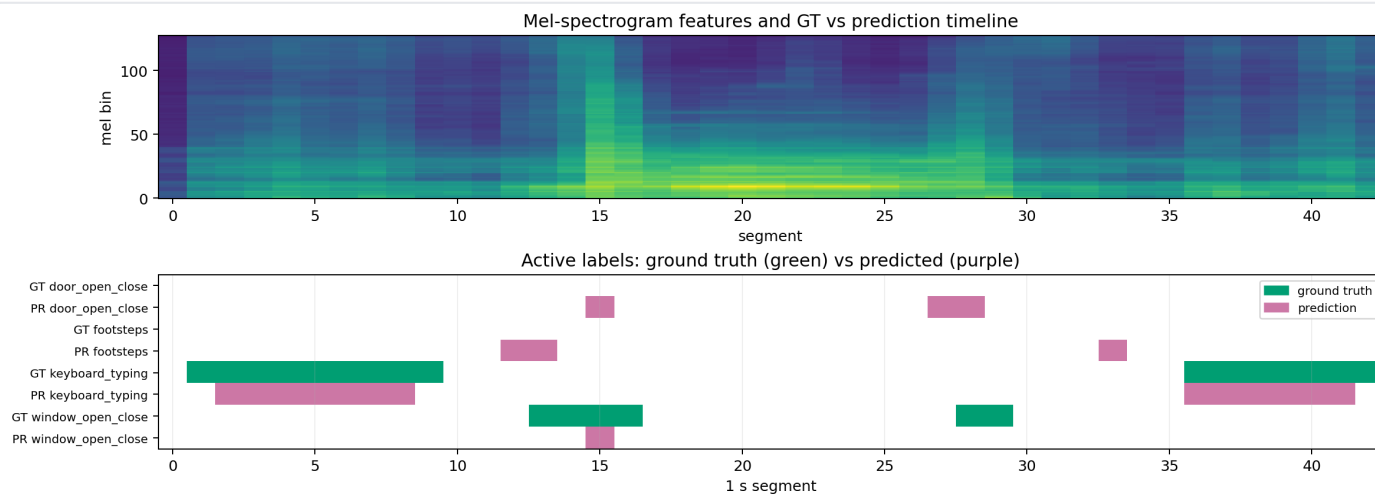
A median filter removes brief gaps and isolated activations by considering neighbouring segment decisions.

0.541 → **0.559**

Window size 5 was the optimum: +1.8 Macro-F1 points and 0.655 Micro F1 on the non-hidden test set.

Larger windows oversmooth short events, so the same mechanism that helps stable classes starts to erase transients.

Hypothesis 4: the remaining errors are mainly temporal and acoustic ambiguities



Strong cases

Long, stable events such as **running water**, microwave, and cutlery are detected with close temporal overlap.

Confusions

Window open/close is confused with door events; footsteps may appear as extra activations.

Failure mode

Feature bursts trigger implausible short predictions because LR has no explicit sequence or duration model.

What the investigation changed and what remains

Macro F1

0.559

median-filtered LR, non-hidden test

Best use today

Local monitoring

activity logging and event summaries

Not ready for

Immediate alerts

latency, false detections, and timing errors
remain

Evidence-backed design

- Regularized LR is a large step up from the tree baseline.
- Threshold calibration matters more than blanket class balancing.
- Small-scale smoothing helps; aggressive smoothing harms transients.

Next technical step

- Use shorter overlapping windows or a temporal sequence model.
- Add class-specific duration constraints.
- Evaluate across diverse rooms, microphones, and users before automation.